

Phone capture → measured floor plan + damage report

Atif Husain · Applied AI Engineer case study · repo: github.com/MA-Husain/floor_plan_pipeline

1. What it does

One command turns a phone recording of a home into a **stitched floor plan**: every room's length, width, area and ceiling height, doors and windows, which rooms connect, **damage** on walls and ceilings (crack, stain, mould, peeling, spalling) with its size in m², the hidden problem it suggests and a repair line — and a **95 % error margin on every number**. Three inputs, same output:

Recording	Phone	Accuracy (measured)
LiDAR scan (Stray Scanner app)	iPhone 12 Pro or newer Pro	room sizes ± 2 cm (95 % interval)
Video (Camera app)	any iPhone 15+	1.4–3.3 % of LiDAR on a slow walk; much worse on fast walks
Photos , one folder per room	any iPhone 15+	4–30 % per room, with wide (honest) margins

2. How to capture (full one-page protocol: docs/CAPTURE_PROTOCOL.md)

Before	All lights on, curtains open, internal doors fully open, lens wiped. Mirrors/glass/wet floors are fine — never stand still facing one.
LiDAR	Install Stray Scanner (free, App Store), defaults. Start by the front door, phone upright at chest height , record. In each room: walk along the walls ~1 step/s, one slow full turn (~8 s), tilt up to the ceiling (count 1-2-3) and down to the floor once, point at each door/window for a second. Walk through the middle of every doorway. Finish where you started. ~1 min per room, under 6 min.
Video	Camera app → Video, 1x lens, upright. Same walk as LiDAR. Turn slowly — one full turn ≈ 8 s. No zoom.
Photos	Camera app → Photo, 1x, upright. Per room 4–8 photos standing inside it: one from each corner + one per doorway looking into the next room (this joins rooms into one plan). Small rooms: 2–3 photos.
Hand over	AirDrop to the computer: the Stray folder, or the .MOV, or photos as <code>my_flat/Kitchen/</code> , <code>my_flat/Bedroom1/</code> , ... (one folder per room).

3. Install, run, open the results

```
# once (needs git + Python 3.12; ~8 GB of model weights, then fully offline)
git clone https://github.com/MA-Husain/floor_plan_pipeline.git && cd floor_plan_pipeline
python3.12 -m venv venv && source venv/bin/activate
pip install -r requirements.txt
python scripts/fetch_weights.py

# every capture: one command, the type is detected automatically
python run_capture.py <Stray folder | video.MOV | my_flat>
```

Output in outputs/<name>/	Open with	What it shows
plan.png	any image viewer	the floor plan: room names, sizes, ceiling heights; damaged walls marked in red (D1, D2...)
damage.png	any image viewer	evidence: the photo each damage was found in, damaged 0.4 m tiles boxed, class / room / area / rule
plan.json	any text editor / code	every number with its ± margin: rooms, openings, adjacency, damage, concealed-damage flags, repair items, error model
terminal	—	a damage summary is printed at the end of each run

Time on an Apple M5 laptop: LiDAR ~7 min, photos ~7 min, video ~10–15 min per minute of footage.

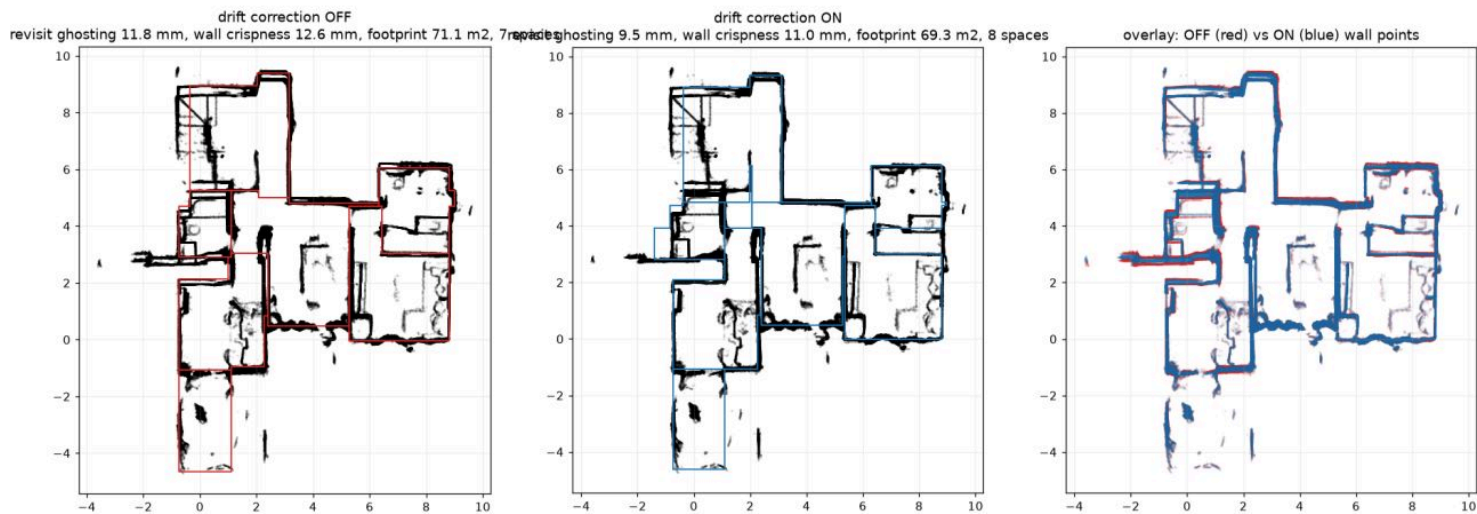
4. Results on the provided sample data



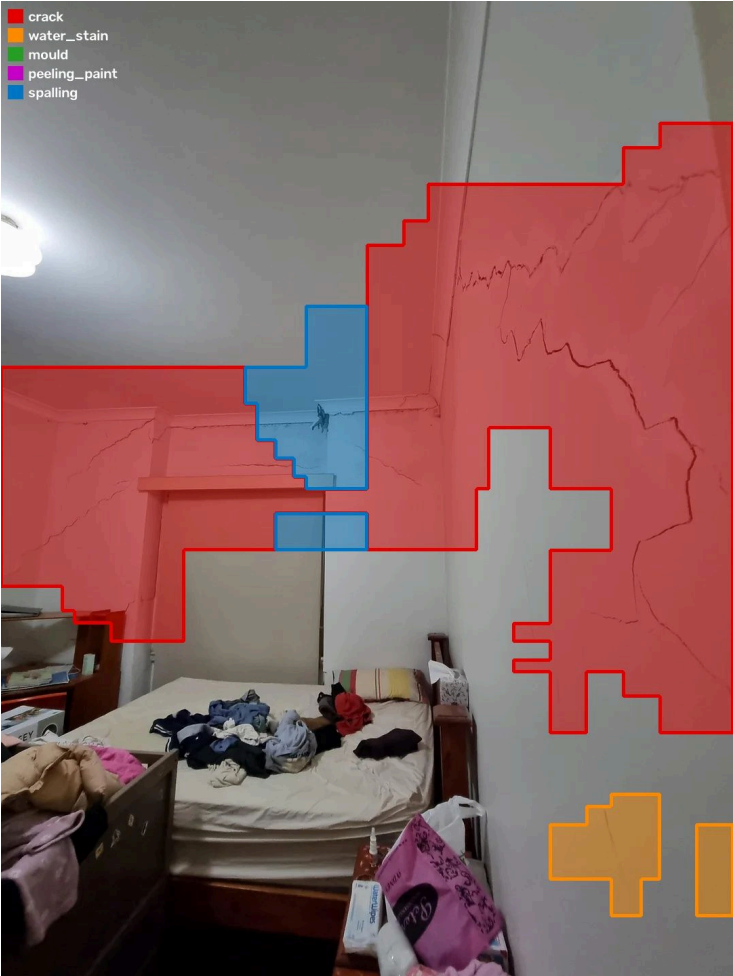
LiDAR floor plan, capture c7d28f72c6 (held out from all tuning). Ceilings shown only where actually seen (2.3–2.4 m = a dropped ceiling over the

wet rooms). D1 = flagged wall.

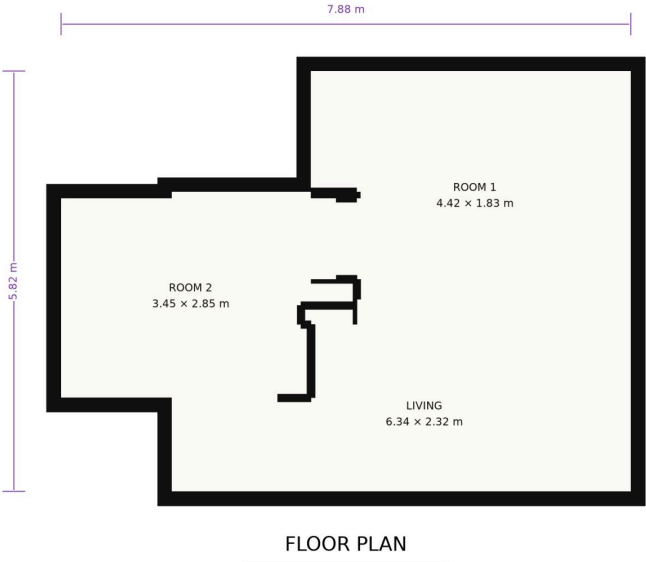
Drift ablation - c7d28f72c6 (ghosting = median gap between two sightings of the same wall ≥ 20 s apart; crispness = median distance of wall points to their plane; lower is better)



Drift on/off (required ablation): correcting the phone's accumulated position drift reduces doubled walls from 11.8 to 9.5 mm.



Damage on a real cracked bedroom photo: wall and cornice cracks found (red), spalling (blue), furniture ignored.



Video-only plan of the single-room sample (no LiDAR used): overall 7.88 × 5.82 m vs LiDAR 7.99 × 6.02 m.

5. Scorecard (details: docs/BENCHMARK.md, docs/COMPLIANCE_MATRIX.md)

Requirement	Result
Three input tiers, one output, one command	Done
Room segmentation (held-out capture)	6/6 rooms, 0 splits/merges, types 6/6
Drift accountability with on/off ablation	Done (11.8 → 9.5 mm, 56.8 → 50.9 mm)
Damage regions + concealed-damage rules + scope	Done — classifier 99.4 % damaged-vs-clean on public BD3 test set
Margin of error on every measurement	Done (tier-specific error budget)
Repeatability ≤ 1 cm (same rooms, two captures)	Fail: 17.2 cm — captures pick different surfaces (wardrobe front vs wall)
Photo tier ±8 % / video ±3 %	Partial: video 1.4–3.3 % on a slow walk, 35–60 % on fast ones; photos 4–30 %
Head-to-head vs magicplan (2 rooms, tape truth)	No iPhone Pro available → compared at photo tier: ours better on 1 of 4 dimensions
Fix loop (declared before the fix, before/after)	Shipped: 20.3 → 17.2 cm; predicted ~4 cm (missed) — post-mortem in the report

Models (all local, disclosed): MoGe-2 (metric depth, chosen by measuring vs LiDAR: 0.955×), MapAnything (camera poses), CLIP + a probe trained on BD3 (damage), YOLO-World (fixtures for room typing). **Known limits:** fast handheld video, rooms without doorway photos, small damage spots, mirrors in video/photos.